

Lecture Notes 1

Measuring the Macroeconomy

Intermediate Macroeconomics, EC2211

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Contents and literature

- Measuring the macroeconomy
- Mathematical preliminaries: logarithms and growth rates
- Growth facts

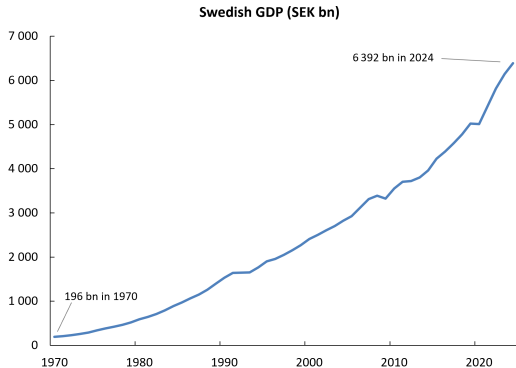
Literature:

- Jones (2024), chapters 1-3 and 6.5

Measuring the macroeconomy

What is GDP and what does it measure?

- Consider this graph
- What do we need to know to make it "interesting"?



Current prices. Source: Statistics Sweden.

GDP = Gross Domestic Product

The value of all final goods and services produced in an economy over a certain period

- National accounts, developed since 1930's, international standards in the [United Nation's System of National Accounts](#)
- Three approaches: expenditure = income = production

Expenditure approach

The National Income Identity

$$Y = C + I + G + EX - IM$$

Y = GDP

C = consumption

I = investment

G = government purchases

EX = exports

IM = imports

NX = $EX - IM$ = net exports = "trade balance"

Example 1

- Homer and Marge grow 100 oranges
- ... sell to Edna, \$0.5 each
- If Edna eats all oranges:
 $Y = C = \$50$
- If Edna eats 90 oranges and save 10 for next period:
 - $C = \$45$ and $I = \$5$ (the increase in inventory is counted as an investment)
 - Again $Y = C + I = \$50$

Example 2

- Homer and Marge grow 100 oranges
- ... sell to Moe, \$0.3 each
- ... Moe sells to Edna, \$0.5 each
- Only *final* expenditure counts, so $Y = C = \$50$
- (Alternatively, you can think of Moe as first making a positive investment when buying oranges, then disinvesting when selling the oranges)

GDP: expenditure approach (ESA2010) by type of use, current prices, SEK million by type of use and year

 Official Statistics of Sweden

Result

[Tools](#) [Pivot manual](#) [Pivot clockwise](#) [Pivot counterclockwise](#)

	2023	
GDP at market prices	6 143 187	← GDP (Y)
Imports of goods and services	3 220 494	← Imports (M)
Household consumption expenditure, incl. NPISH	2 798 083	← Consumption (C)
General government final consumption expenditure	1 622 880	
Gross capital formation	1 551 932	← Investment + government investment
Central government, fixed capital formation	165 117	← Government purchases (G)
Local government gross fixed capital formation	146 591	← Government purchases (G)
Exports of goods and services	3 390 786	← Exports (X)

Expenditure approach to Swedish GDP in 2023

- Consumption here refers to household consumption
- Investment here refers to private investment...
- ... and includes changes in inventories

	SEK bn	% of GDP
Gross domestic product	6,143	100
Consumption	2,798	46
Investment	1,241	20
Government purchases	1,935	31
Net exports	170	3
. <i>exports</i>	3,391	55
. <i>imports</i>	3,220	52

Source: [Statistics Sweden](#)

Income approach

$$\text{GDP} = \text{"wages"} + \text{"profits"}$$

- "Wages" = compensation to workers
- "Profits" = compensation to capital owners (interest payments, dividends)

Example 1

- Homer and Marge grow 100 oranges
- ... sell to Edna, \$0.5 each
- $\text{GDP} = \text{Homer and Marge's income} = \50

Example 2

- Homer and Marge grow 100 oranges
- ... sell to Moe, \$0.3 each
- ... Moe sells to Edna, \$0.5 each
- $\text{GDP} = \text{Homer and Marge's income} + \text{Moe's profit} = \$30 + \$20 = \50

Income approach to Swedish GDP in 2023

	SEK bn	% of GDP
Gross domestic product	6,143	100
Compensation of employees	2,938	48
. <i>wages and salaries</i>	2,426	39
. <i>employers' social contributions</i>	511	8
Operating surplus and mixed income	1,980	32
Taxes on production and imports less subsidies	1,225	20

Source: [Eurostat](#)

Note: The last row ("Taxes ...") consists mostly of VAT, import tariffs and payroll taxes. Income taxes and corporate profit taxes are *not* in that item.

Production approach

GDP = sum of value added in production

- Only "new value" generated in each production step is counted
- Consider example 2:
 - Homer and Marge generate \$30 worth of value when they grow apples
 - Moe generates an additional \$20 worth of value, maybe by advertising, packaging, bringing oranges closer to customers etc

Production approach to Swedish GDP in 2023

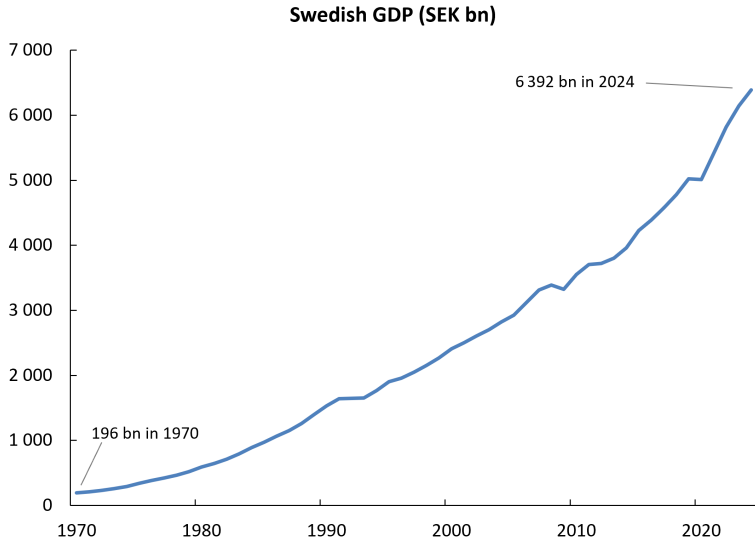
	SEK bn	% of GDP
Gross domestic product	6,143	100
Agriculture, forestry and fishing	79	1
Mineral extract	38	1
Manufacturing	823	13
Electricity, gas, steam and air conditioning, water supply, waste	173	3
Construction	354	6
Wholesale and retail trade	559	9
Transport and storage	206	3
Hotels and restaurants	93	2
Publishing, media, sound, telecom, programming, IT	397	6
Financial services and insurance activities	246	4
Real estate activities	476	7
Legal, accounting and activities of head offices, management consultancy	175	3
Architectural and engineering activities, R&D	138	2
Advertising and market research, veterinary activities	58	1
Administrative and support service activities	211	3
Education	53	1
Human health, residential care, social work	132	2
Arts, entertainment, recreation and other service activities	86	1
Non-profit institutions servicing households	67	1
Government	1,123	18
Taxes on products less subsidies	655	11

Source: [Statistics Sweden](#)

Note: The last row ("Taxes ...") consists mostly of VAT and import tariffs.

Measuring the macroeconomy: prices and quantities

Do we produce more today or are prices just higher?



Prices and quantities: preliminaries

- Suppose that just two goods are produced: apples and oranges
- Let Q denote quantity and P price
- In year 1, $Q_1^A = Q_1^O = 100$ and $P_1^A = P_1^O = 1$
- In year 2, $Q_1^A = 100$, $Q_1^O = 150$, $P_1^A = 2$ and $P_1^O = 1$

- *Nominal* GDP is then:

$$Y_1^N = 1 \times 100 + 1 \times 100 = 200$$

$$Y_2^N = 2 \times 100 + 1 \times 150 = 350$$

- How much more do we produce in period 2? How much have prices increased?

Prices and quantities: preliminaries

- Use prices from period 1 to calculate "real" GDP:

$$Y_1^{P_1} = 1 \times 100 + 1 \times 100 = 200$$

$$Y_2^{P_1} = 1 \times 100 + 1 \times 150 = 250$$

- Use prices from period 2 to calculate "real" GDP in year 1:

$$Y_1^{P_2} = 2 \times 100 + 1 \times 100 = 300$$

$$Y_2^{P_2} = 2 \times 100 + 1 \times 150 = 350$$

- The growth rate of real GDP is then

$$\frac{Y_2^{P_1} - Y_1^{P_1}}{Y_1^{P_1}} = 25\% \text{ if prices from period 1 are used}$$

$$\frac{Y_2^{P_2} - Y_1^{P_2}}{Y_1^{P_2}} = 17\% \text{ if prices from period 2 are used}$$

Prices and quantities: preliminaries

- Not obvious in this simple example how to separate quantities from prices
- These two methods are the *Laspeyres* and *Paasche* index methods, respectively
- In the real world, things are much more complicated
- Measures are not exact, methods vary between countries!

How do we compare "today" with "yesterday"?



Prices and quantities

- Difficult to separate price changes from quality improvements
- Consumption baskets change over time
 - Our preferences change
 - New goods and services are developed
- Problems are more salient when comparison is over long time periods or if price movements are large
- A typical method used by statistical agencies is the *chain weighted* index:
 - "Average" of Laspeyres and Paasche
 - Chain-linked over short time-intervals

Nominal and real GDP: example terminology

- Nominal GDP: "current prices, million SEK"
- Real GDP:
 - "2015 prices, million SEK"
 - "Constant prices, reference year 2024, SEK million"
 - "Chain linked volumes, index 2010=100"
 - "Chain linked volumes (2010), million euro"

Price indices: some examples

- GDP deflator:
 - $\text{GDP deflator} = \text{nominal GDP} / \text{real GDP}$
 - Average price of goods produced in the economy, weighted by importance in production
- CPI, "Consumer price index":
 - Average price of goods consumed in the economy, weighted by importance in households' consumption baskets
- The main difference between these relates to international trade
 - Many goods in the consumption basket are imported

From 1970 to 2023

GDP increased from SEK 196 bn in 1970 to SEK 6,143 bn in 2023:

$$\frac{Y_{2023}^N}{Y_{1970}^N} = \frac{6,143}{196} \approx 31.4$$

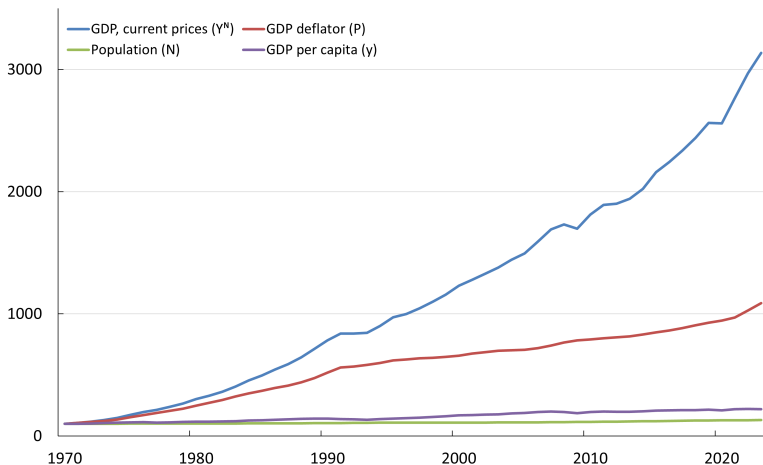
In addition to separating prices from quantities, we often also want to account for population growth:

$$\frac{Y_{2023}^N}{Y_{1970}^N} = \frac{P_{2023} N_{2023} y_{2023}}{P_{1970} N_{1970} y_{1970}} = \frac{P_{2023}}{P_{1970}} \frac{N_{2023}}{N_{1970}} \frac{y_{2023}}{y_{1970}}$$

where y is real GDP per person and N the population size

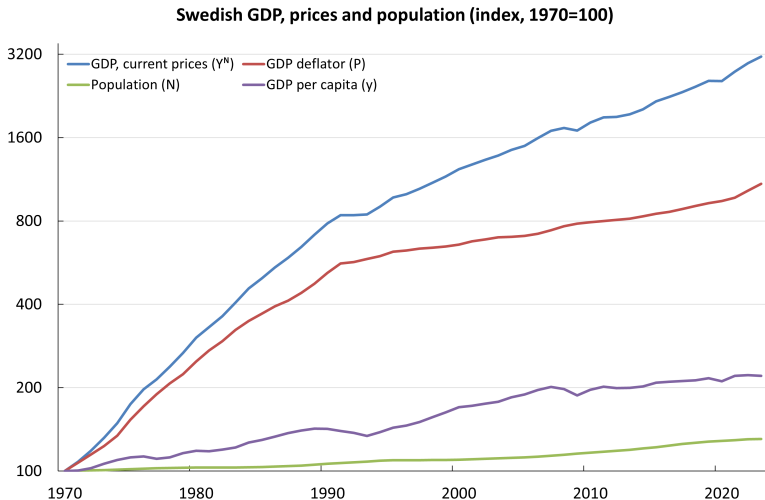
From 1970 to 2023

Swedish GDP, prices and population (index, 1970=100)



$$\frac{P_{2023}}{P_{1970}} \approx 10.9, \frac{N_{2023}}{N_{1970}} \approx 1.3 \text{ and } \frac{Y_{2023}}{Y_{1970}} \approx 2.2$$

From 1970 to 2023



This is the same graph, but now using a log scale for the y-axis

Why logarithms?

In a graph with a log scale for the vertical axis:

- Each vertical step represents the same *relative* change
- A straight line means that the *growth rate* is constant

The previous graph indicates that:

- The growth rates of prices and nominal GDP were high from 1970 to 1990
- There is no clear long-run trends in the growth rates of real GDP per capita or the population

Mathematical preliminaries: logarithms and growth rates

Mathematical preliminaries: the natural logarithm

Economists use both $\ln x$ and $\log x$ to denote the natural logarithm of x . By definition:

$$x = e^a \Leftrightarrow a = \ln x$$

Properties:

$$\ln(xy) = \ln x + \ln y$$

$$\ln(x/y) = \ln x - \ln y$$

$$\ln x^\alpha = \alpha \ln x$$

and if x is small:

$$\ln(1+x) \approx x$$

$$\ln(1+x)^\alpha \approx \alpha x$$

Growth rates

Economists are often interested in growth rates. Suppose that y grows at rate g , that is:

$$y_{t+1} = (1 + g_{t+1})y_t$$

Note from the properties of the natural logarithm that we then have:

$$\ln y_{t+1} = \ln [(1 + g_{t+1})y_t] = \ln (1 + g_{t+1}) + \ln y_t \approx g_{t+1} + \ln y_t$$

All this shows different ways that we can express the growth rate:

$$g_{t+1} = \frac{y_{t+1} - y_t}{y_t} \approx \ln y_{t+1} - \ln y_t = \ln \left(\frac{y_{t+1}}{y_t} \right)$$

The constant growth rule

Suppose that the growth rate g is constant so that, for all t ,

$$y_{t+1} = (1 + g)y_t$$

Suppose that we start at $t = 0$, with y_0 , and that there are 3 periods:

$$y_1 = (1 + g)y_0$$

$$y_2 = (1 + g)y_1$$

$$y_3 = (1 + g)y_2$$

The constant growth rule

Substituting recursively:

$$\begin{aligned}y_3 &= (1 + g)y_2 \\&= (1 + g)\underbrace{(1 + g)y_1}_{y_2} \\&= (1 + g)(1 + g)\underbrace{(1 + g)y_0}_{y_1} \\&= (1 + g)^3 y_0\end{aligned}$$

The constant growth rule:

$$y_t = (1 + g)^t y_0$$

Measuring the macroeconomy: exchange rates and international comparisons

How do we compare GDP across countries?

- GDP in 2019: 95.1 trillion yuan in China, \$20.6 trillion in the United States
- Exchange rate 1 USD = 6.91 yuan
 - Convert Chinese GDP to dollars at market exchange rates:

$$95.1 \text{ trillion yuan} \times \frac{\$1}{6.91 \text{ yuan}} = \$13.8 \text{ trillion}$$

- But the price level in China was around 68.4 percent of the price level in the United States
 - Convert Chinese GDP to dollars using U.S. prices (i.e. at **purchasing power parity**)

$$95.1 \text{ trillion yuan} \times \frac{\$1}{6.91 \text{ yuan} \times 0.684} = \$20.1 \text{ trillion}$$

- So PPP adjusted GDP in China is roughly as high as that in the United States

Growth facts

"It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts."

-Sherlock Holmes (A Scandal in Bohemia, 1891)

Growth facts

- There is enormous variation in GDP per capita across countries
- Growth rates vary substantially across countries
- Growth rates also vary over time. Sustained growth is a recent phenomenon.

Fact 1: There is enormous variation in GDP per capita across countries

Country	GDP per capita	Percent of U.S.
United States	62,589	100
Sweden	52,433	83.8
Germany	51,191	81.8
Japan	39,704	63.4
Mexico	18,737	29.9
China	14,129	22.6
India	6,711	10.7
Bangladesh	4,658	7.4
Chad	1,615	2.6
Burundi	790	1.3

GDP per capita in 2019, in 2017 U.S. dollars. Source: PWT 10.01.

Fact 2: Growth rates vary substantially across countries

	Percent change in GDP per capita since		Annual growth rate (%)
	1960	1990	1960-2019
United States	228	58	2.0
Sweden	289	73	2.3
Germany	359	82	2.6
Japan	625	35	3.4
Mexico	181	51	1.8
China	1,321	482	4.6
India	475	369	3.0
Bangladesh	198	204	0.1
Chad	12	35	0.2
Burundi	5	-22	0.1

Last year of data is 2019. Calculations based on GDP per capita in 2017 U.S. dollars. Source: PWT 10.01.

Fact 3: Growth rates vary over time

	Annual growth rate (%)	
	1960-1990	1990-2019
United States	2.5	1.6
Sweden	2.7	1.9
Germany	3.1	2.1
Japan	5.8	1.0
China	3.0	6.3
Mexico	2.1	1.4
India	0.7	5.5
Bangladesh	-0.1	3.9
Chad	-0.6	1.0
Burundi	1.0	-0.9

Calculations based on GDP per capita in 2017 U.S. dollars. Source: PWT 10.01.

Fact 3: Growth rates vary over time

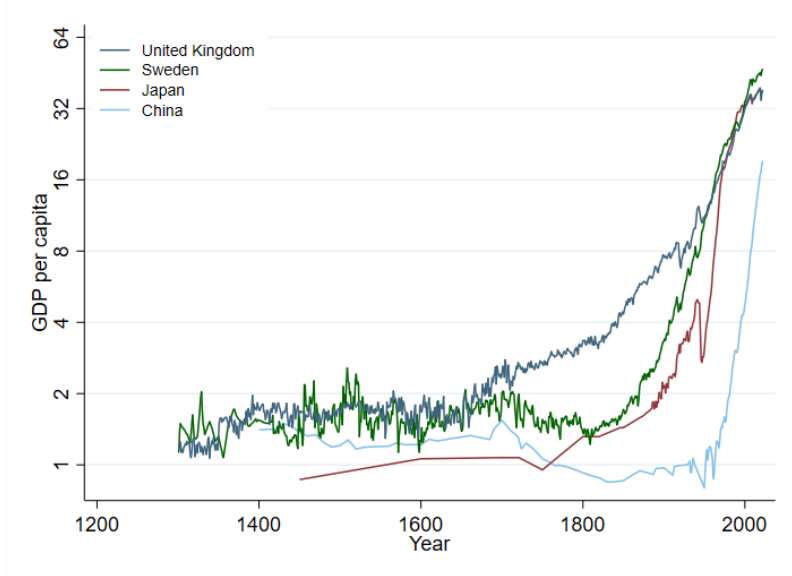
Very little growth before 1800s

- Lack of good data
- Best estimates:
 - ... no growth in first millennium A.D.
 - ... then less than 0.1 percent per year to year 1800

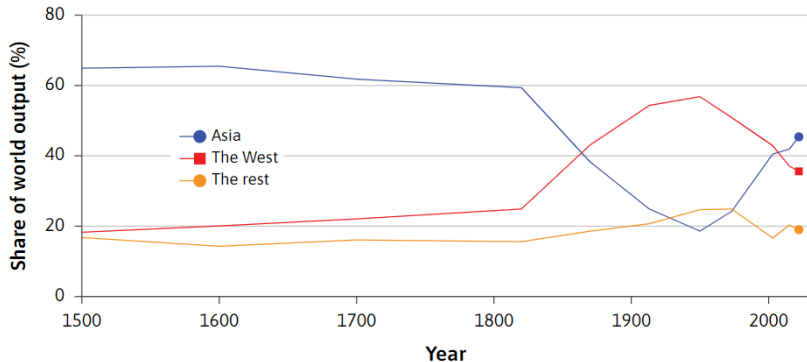
Rapid growth thereafter

- U.S. GDP per capita has increased by factor 20 since year 1800
- ... but not everywhere
- ... and starting dates vary

Fact 3: Growth rates vary over time

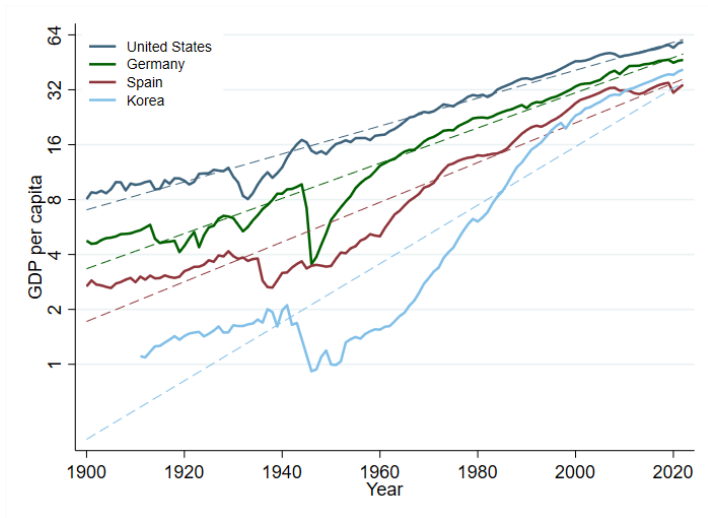


Rapid growth (both population and productivity) in "the West" from 1800s



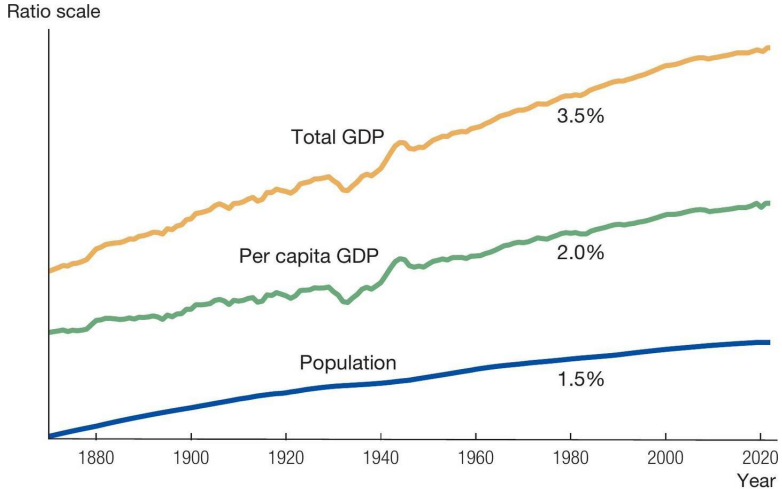
Different regions' share of world output. Source: [COREecon Microeconomics](#), figure 1.18

Growth rates vary over time, but relatively stable just below 2 percent per year in the United States



Real GDP per capita, 2017 USD 1000s, and trend lines. Source: [Maddison project](#)

U.S. population, GDP, and GDP per capita



Sources: Angus Maddison, “Statistics on World Population, GDP and Per Capita GDP, 1 AD–2006 AD,” and the Bureau of Economic Analysis.

Figure 3.9 in Jones (2024)

The great divergence

- Cross-country differences in GDP per capita were relatively modest until 1700s
- Emergence of large cross-country differences in living standards from 1700 onwards
- But less so in recent decades
 - In 1990, 58 percent of the world's population lived in low-income countries
 - ... today, only 9 percent (source: Gapminder)

The distribution of world population by GDP per capita

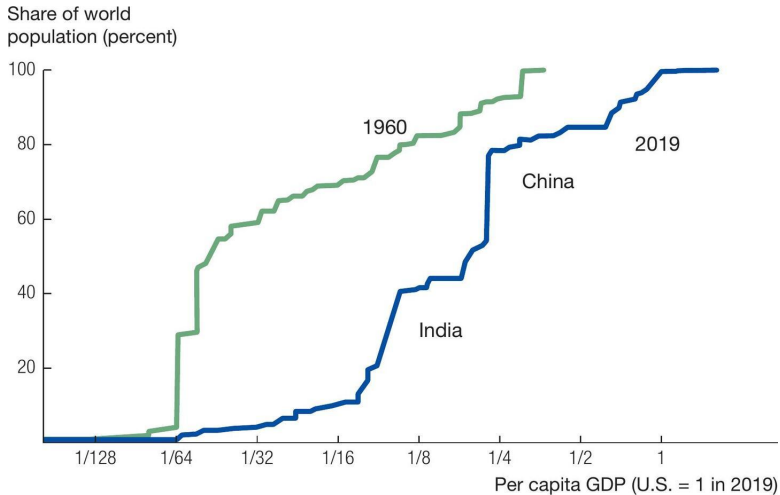


Figure 3.8 in Jones

The share of the world population living in extreme poverty has fallen substantially

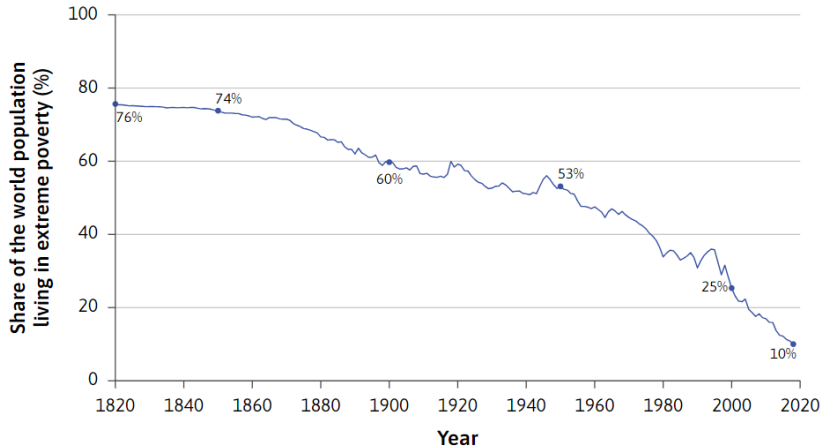
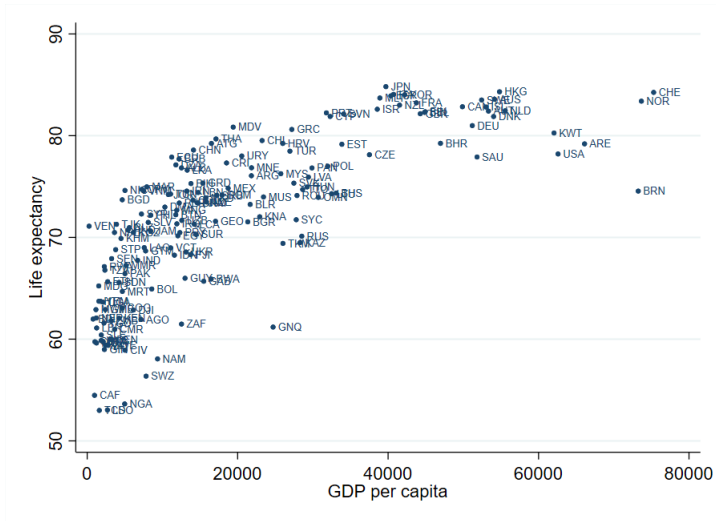


Figure 1.3 in COREecon Microeconomics (2024)

GDP is not welfare, but ...



Life expectancy at birth (2023 Human Development Index) and GDP per capita (PWT 10.01). Ireland and Qatar were dropped from the sample.
A more entertaining and dynamic version of this graph: [Hans Rosling: 200 years in four minutes](#)

Growth since 1960

- Some major economies have maintained or improved their position relative to the US (e.g. France, Italy, Spain and Japan)
- In the poorer group, some have improved (Korea, India, China) while others have fallen further behind (Nigeria, Ethiopia)
- Great heterogeneity: growth miracles and growth disasters abundant

Questions

- Why are some countries so rich and others so poor?
- Why do some countries grow rapidly and some stagnate?
- Can growth continue in the long run? What are the sources and limitations of growth?
 - Population growth?
 - Physical capital?
 - Education, research, ideas?

Malthus → Solow → Romer

What we did

- Measuring the economy: national accounts; real and nominal GDP; price indices
- Mathematical preliminaries: logarithms and growth rates
- Growth facts

Next time: the production function; growth accounting, the Solow model